

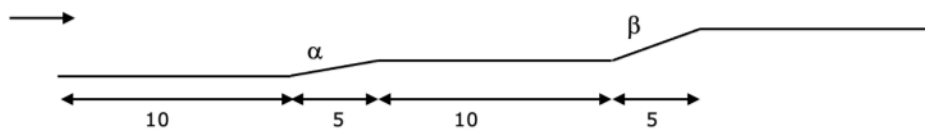
Applied CFD Project



Name:..... Dominik Balasko
 Matriculation Number:.....
 E-mail:..... dominik.balasko@tum.de
 Moodle Group:.....

Name:.....
 Matriculation Number:.....
 E-mail:.....
 Moodle Group:.....

Investigated geometry: Ramp geometry with $\alpha = 8^\circ$ and $\beta = 22^\circ$



Solver Setup

Solver Settings	
Advection scheme	High resolution
Timestep control	Physical Timescale, Timestep size: 1×10^{-4} s
Fluid properties	Air as ideal gas, $p_{ref} = 0$ [atm]
Fluid model	Continuous fluid, non-buoyant

Boundary Conditions	
Inlet	Supersonic inlet with total pressure, static pressure and total temperature with medium turbulence
Outlet	Supersonic outlet
Walls	Non-slip smooth walls
Symmetry	Applied where appropriate

Grid Properties			
Property	Uniform	Refined	Refined with extra prisms
Grid size [cells]	77 161	60 289	70 039
Quality (min)	0.927	0.269	0.22
Skew value (min)	0.76	0.18	0.18

1 Numerical Setup

1.1 Fluid Model

At the investigation of the flowfield around double ramp geometry 3 simulations were performed. One inviscid case, and two viscous simulations using $k - \varepsilon$ and *BSL EARSM* turbulence model. The material was chosen as Air Ideal Gas for the viscous cases. This material is predefined in ANSYS CFX material library. However for the inviscid case a new artificial material had to be defined to exclude any viscous effects from the simulations. The material properties of the Inviscid Air Ideal Gas can be seen in table 1.

Quantity	Symbol	Value	Units	Notes
Specific gas constant	R	287.050	J/(kg K)	Perfect gas EOS $p = \rho RT$
Ratio of specific heats	γ	1.400	–	Calorically ideal
Specific heat at const. p	c_p	1004.500	J/(kg K)	$c_p = \gamma R / (\gamma - 1)$.
Specific heat at const. v	c_v	717.500	J/(kg K)	$c_v = c_p - R$.
Molar mass (ref.)	M	28.965	g/mol	For converting $R = \bar{R}/M$.
Dynamic viscosity (Euler)	μ	0.000	Pa s	Inviscid assumption
Thermal conductivity (Euler)	k	0.000	W/(m K)	No heat conduction

Table 1: Material properties used for air as inviscid ideal gas

1.2 Boundary Conditions

In all three simulation cases the model was defined using Dirichlet Boundary Condition at the inlet and Neumann Boundary Condition at the outlet. The side walls and the top of the domain were set to symmetric. The wall was treated as Free Slip wall for the inviscid case and No Slip Wall with smooth roughness for the viscous simulations. The Inlet Boundary was defined using static pressure (p), total pressure (p_0) and total temperature (T_0). The turbulence intensity was set to medium (5%). The static pressure was used instead of the velocity definition due to the stability of the simulation. The value for static pressure was calculated using the isentropic relation:

Total-to-static pressure relation:

$$p_0 = p \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}} \implies p = \frac{p_0}{\left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}}}$$

Given $p_0 = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$, $\gamma = 1.4$, $M = 3.8$:

$$p \approx 866 \text{ Pa}$$

In Supersonic flow no information travels upstream, so no boundary condition is mathematically required on the outlet. For this reason a zero-gradient Neumann Boundary condition was used there. The summary of the Boundary conditions can be seen in table 2.

Boundary	Type	Values
Inlet	Supersonic	$p_0 = 1 \text{ bar}$, $p = 866 \text{ Pa}$, $T_0 = 900 \text{ K}$, $I_{\text{turb}} = 0.05$
Outlet	Supersonic	Zero-gradient
Ramp Wall	Adiabatic, no-slip / slip (depending on case)	–
Side Wall, Top	Symmetry	–

Table 2: Boundary conditions used in the simulations.

1.3 Discretization

At the investigated supersonic flow scenario shocks with high gradients were expected, which can be critical for a second order spatial discretization scheme. However ANSYS CFX has the High Resolution advection scheme in the feature list, which is a blend between first- and second order discretization. In smooth regions the solver uses second order method for better accuracy and lower numerical diffusion, however at steep gradients (shocks) it can switch to first order upwind scheme for robustness and stability. In every performed simulation the High Resolution advection scheme was used.

The viscous simulations required also a discretization for the turbulence numerics. At this point first order discretization was used due to the experienced instability and oscillations with the High Resolution scheme. The first order scheme helps to damp the oscillations near shocks and helps to stabilize the simulation.

1.4 Simulation Cases

Three cases were simulated: inviscid case and viscous simulation with k-epsilon and BSL EARSIM model.

The inviscid simulation neglects every viscous effect and it solves the Euler-equations. This simulation is expected to converge fast because of the highly simplified inviscid transfer equations. The inviscid simulation can predict the expected shock angle and the characteristic of the flow (driven by shocks and high gradients); however, an inviscid simulation is not a real flow scenario, since in reality viscosity plays an important role. For this reason the flowfield close to the wall can not be predicted using the inviscid simulation method and a more complex viscous simulation is required.

The first viscous simulation was performed using the $k - \varepsilon$ turbulence model. This model is a 2-equation model and uses two extra transport equations, one for Turbulent Kinetic Energy (k) and one for the dissipation rate (ε). The non closed RST is solved using the Eddy-viscosity approach, where turbulence is modeled as viscosity. This approach

suggests isotropy, therefore the $k - \varepsilon$ has stability issues in anisotropic flows and close to walls. For this reason the model is often used together with wall function. In the present $k - \varepsilon$ simulations the $Y+$ value between 30 and 300 was aimed to stay in the log region and be able to use wall function. The model is expected to be robust, but the residuals are expected higher than the inviscid case and due to the more complex modeling approach a slower convergence is expected. This model can already give a more realistic result compared to the inviscid case, especially at the Shock-Boundary layer interaction, however in case of high Mach number and strong shocks a more precise model has to be used, which can handle anisotropy in the flow.

The BSL EARSM turbulence model can handle the anisotropy of the Reynolds Stress tensor, therefore this model is capable of fully resolving the boundary layer and handle the strong shocks. For this reason in this simulation the $Y+$ was kept below 5, which means that the first cell center close to the wall has to be in the viscous sublayer. The modeling approach at every Reynolds Stress model is to use a transport equation for every Reynolds Stress term. This results 6 extra equations. Most RSM also require the ε transport equation, therefore in total 7 extra equations have to be solved. This indicates a lot higher computational effort compared to the 2-equation models. The most accurate results are expected from the BSL EARSM model, due to the capability of handling anisotropy in flows. However these models are also not so robust, therefore poor convergence and higher residuals are expected. A summary of the three modeling approaches can be found in table 3

Model	Viscosity	Turbulence	Purpose
Inviscid (Euler)	No	None	Baseline shock prediction
RANS $k-\varepsilon$	Yes	Isotropic	Shock-boundary layer interaction
RANS BSL-EARSM	Yes	Anisotropic	Improved shock and BL prediction

Table 3: Summary of simulation cases.

2 Grid

The double ramp problem was investigated within a domain with 60 Unit length and 30 Unit height. This domain size avoids any shock reflection from the top of the domain and leaves enough space behind the second ramp for shock interaction.

For the spatial discretization of the domain, a structured grid was employed. This grid type generally exhibits lower numerical diffusion compared to its unstructured counterpart, provided that the mesh is aligned with the main flow direction. Moreover, structured grids inherently eliminate the need for an additional connectivity matrix, leading to reduced memory consumption and improved computational efficiency. However, the generation of a structured grid often requires a careful blocking strategy tailored to the geometry, which makes it less practical for highly complex configurations. As a result, unstructured grids are more commonly adopted in industrial applications, while structured grids are primarily utilized in academic and research-oriented studies.

For the present investigation, two different structured grid was generated: a universal mesh and a refined mesh, enabling a comparative assessment of grid resolution effects on the solution accuracy.

2.1 Universal grid

The inviscid simulation was performed on a universal grid. At this stage, there was no prior information about the flow field; therefore, it was not known where to expect high gradients and where mesh refinement would be required. Consequently, a simple blocking strategy was adopted, consisting of five blocks, created by splitting the blocking at the kinks of the geometry (see Fig. 1).

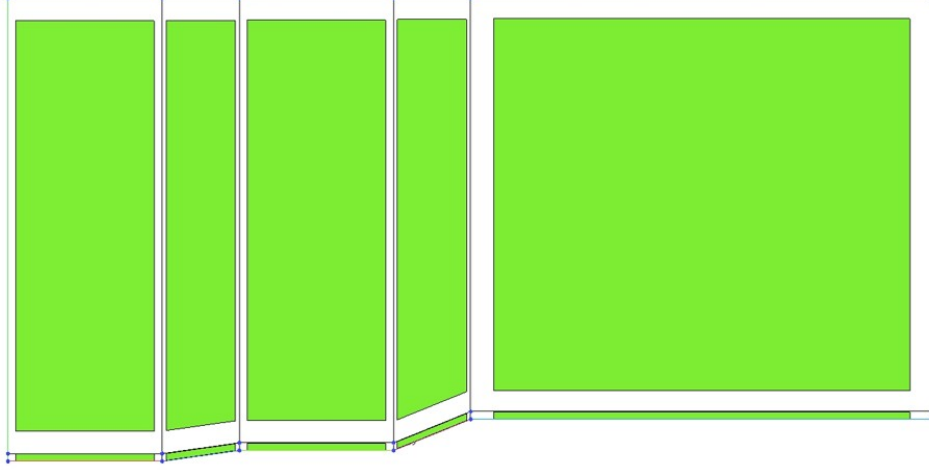


Figure 1: Blocking used for the universal mesh

The inlet was divided into 128 nodes, with the distribution defined using the Exponential 1 function. For the $k-\varepsilon$ simulation, an additional prism layer was introduced to control the Y^+ value at the wall. To achieve this, an extra horizontal split was added to the domain, resulting in the creation of thin boundary layer (BL) blocks. These BL blocks were divided vertically into 25 grid points using the Exponential 1 law with a growth rate of 1.15 (see Fig. 2), which ensured that the Y^+ value remained within the logarithmic region (see fig. 5). Overall this mesh has a minimum cell quality of 0.93, minimum orthogonality of 67.95° and min. skew value of 0.76.

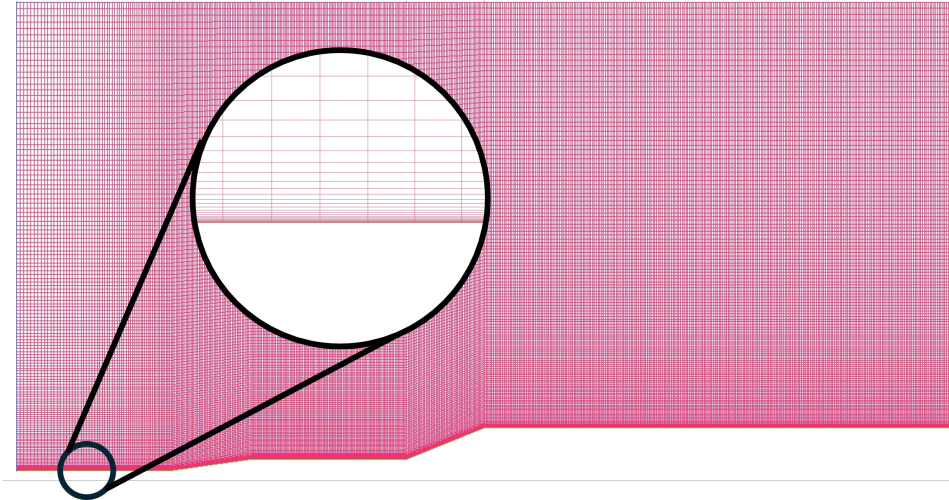


Figure 2: Grid generated for the universal mesh with the prism layers highlighted

2.2 Refined Grid

Based on the results of the universal grid simulation, a mesh refinement was performed in regions with high gradients. To achieve this refinement, two additional splits were introduced in the blocking (see Fig. 3), which enabled a more detailed resolution of the shock formed after the interaction of the two shocks. The angles of the refinement regions were aligned with the shock angles obtained from the previous simulation. As shown by the updated blocking, this approach resulted in highly distorted cells with poor quality (0.22), low skew value (0.18), and low orthogonality (15.9°).

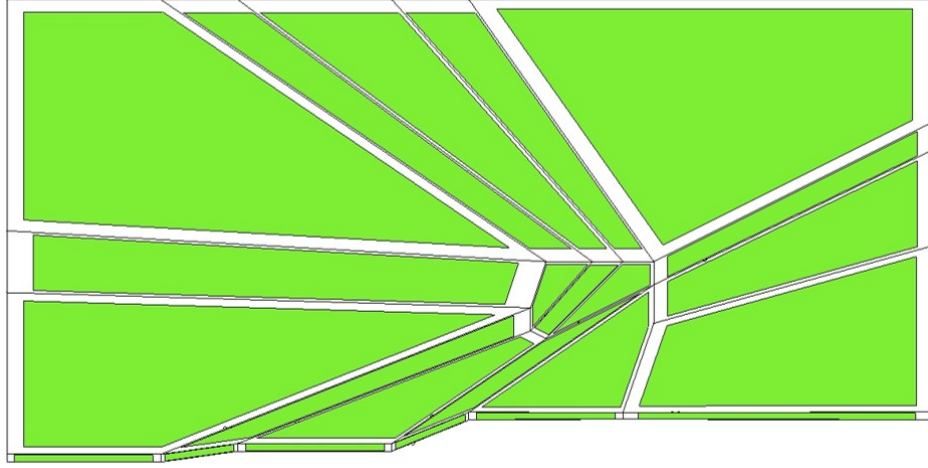


Figure 3: Blocking used for the refined mesh

The refinement regions around the shocks were divided into 32 nodes, with the point distribution defined using the Biexponential law. This allowed for a higher concentration of cells in the middle of the block, where the shock location was expected. At the lower region of the outlet an O-block was introduced to improve mesh quality and avoid cells with excessively high skewness (low skew value).

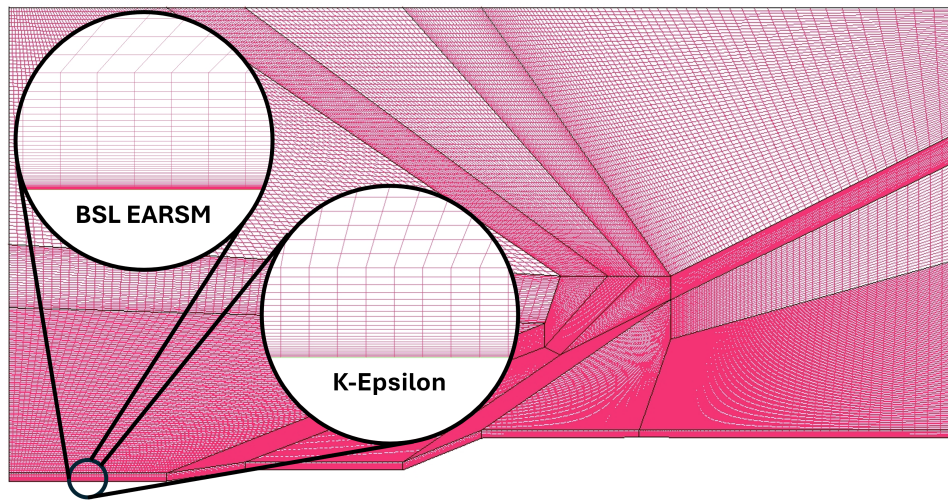


Figure 4: Refined grid with prism layers highlighted

Furthermore, the prism layer was refined for the BSL EARSIM simulation, where 64 nodes were generated within the boundary layer (BL) block (see Fig. 4). This refinement

ensured that the Y^+ value remained below 5 across the entire ramp (see Fig. 5), allowing the boundary layer to be fully resolved.

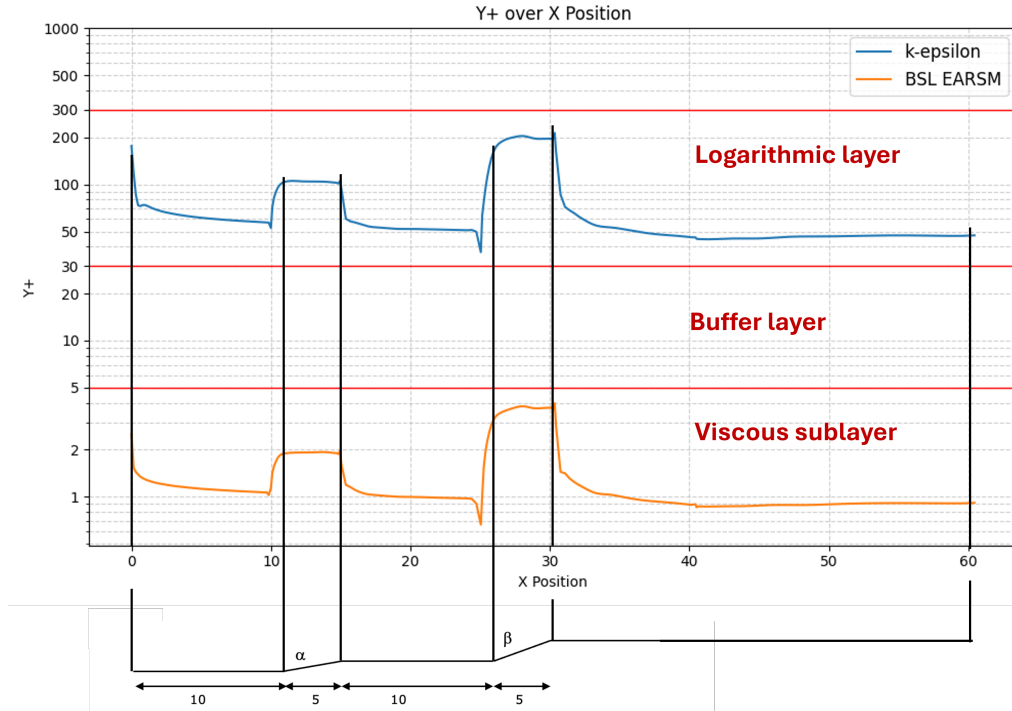


Figure 5: Y^+ plotted over X position

To evaluate whether the generated mesh was suitable for further investigations, the two meshes were compared using the $k-\varepsilon$ turbulence model. Both meshes were simulated under the same flow conditions, and their convergence behavior as well as the resulting flow quality were analyzed (see Convergence and Results).

3 Convergence

In every CFD simulation, the residuals serve as important indicators of the convergence of the solution. Lower residuals imply a smaller deviation between the solved equations and the equations that still need to be satisfied. However, to more accurately monitor the convergence of the flow, a volume probe was placed at the position (50, 10), where the relevant physical quantities were continuously monitored. The convergence limit was set to 10^{-6} in every simulation.

3.1 Residuals

By investigating the Residuals of the performed simulations a clear trend can be seen. The inviscid simulation reached the convergence limit after 218 iterations. Every Residual curve is smooth without oscillations. The small jump in the residuals shows the switch from global timestepping to local timestepping.

The $k-\varepsilon$ simulation on the same grid converged slower, the residual curve already has some oscillations, which can be explained by the poor performance of k-epsilon model in anisotropic flow.

By comparing the Refined mesh with the universal mesh it is visible that the Refined mesh converged slower, however it still reached the convergence limit within 646 iterations compared to the 382 iterations of the universal grid.

The BSL EARSIM model exhibited poor convergence; however, it still reached the convergence criteria, although it required significantly more iterations (1270 iterations) compared to the previous simulations. Some oscillations in the residuals are still visible, but in certain cases, these oscillations are smaller than those observed with the $k-\varepsilon$ model, which confirms the better performance of RSM models in anisotropic flows.

The slow convergence can be explained by the tight coupling between the governing equations. In the case of RSM models, the Reynolds stress tensor—for which six additional transport equations are solved—directly appears in the RANS momentum equations. This strong coupling makes the system stiffer and therefore slows down convergence, particularly in regions with high flow anisotropy.

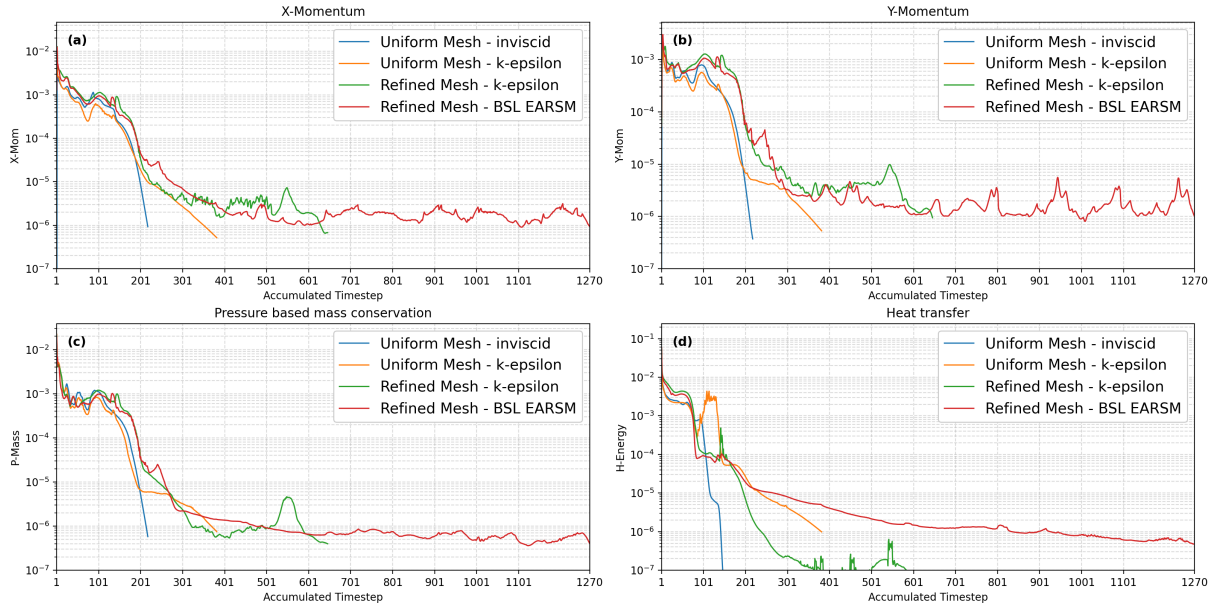


Figure 6: The evolution of Residuals RMS U-Mom (fig. a), RMS V-Mom (fig. b), RMS P-Mass Residual (fig. c) and RMS H-Energy (fig. d) plotted in dependence of timestep

3.2 Convergence of physical quantities (U, p, ρ, T)

To ensure the convergence of the simulations, the velocity (U), pressure (p), density (ρ), and temperature (T) were monitored throughout the iterations. In terms of these physical quantities, there is no significant difference observed between the simulations compared to the residuals. All simulations reach a converged state after approximately 200 iterations. Furthermore, no oscillations or unphysical phenomena are present, confirming that the simulations are fully converged.

4 Results

The main characteristics of the flow field were analyzed using the total pressure distribution (see Fig. 8). Two distinct shocks can be identified. The first shock, originating at

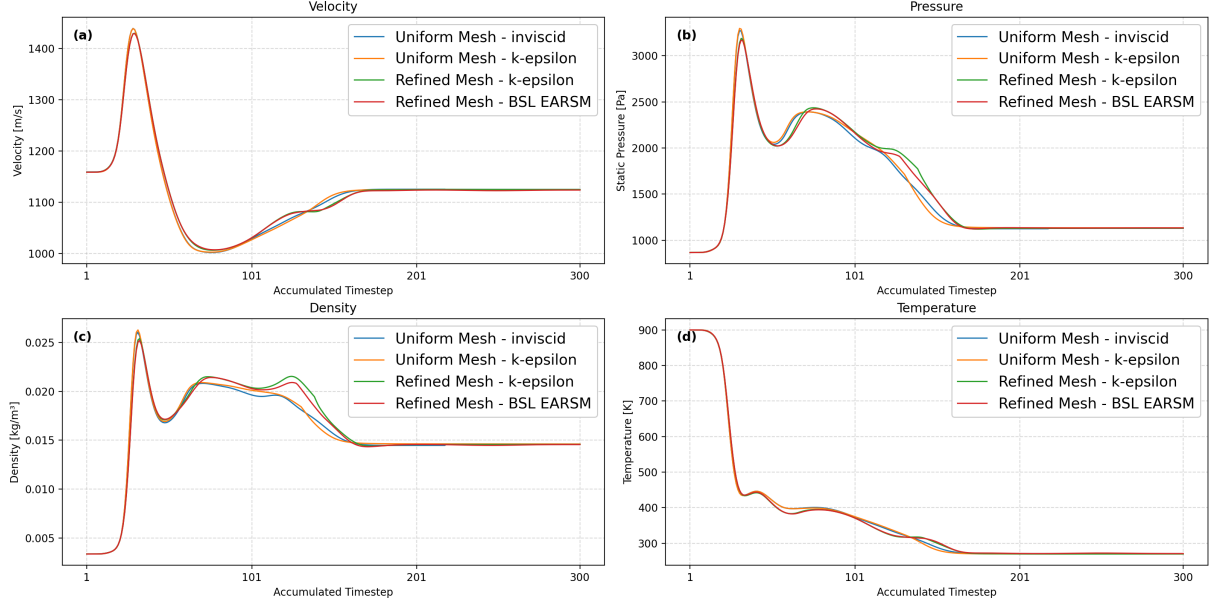


Figure 7: The convergence of *Velocity* (fig. a), *Static pressure* (fig. b), *Density* (fig. c) and *Temperature* (fig. d) plotted in the position (50, 10)

the first ramp, exhibits a weaker strength, as indicated by the moderate total pressure drop across the shock ($\Delta p_{01} \approx 6\,000$ Pa). In contrast, the second shock has a significantly higher strength, with a total pressure drop of $\Delta p_{02} \approx 40\,000$ Pa.

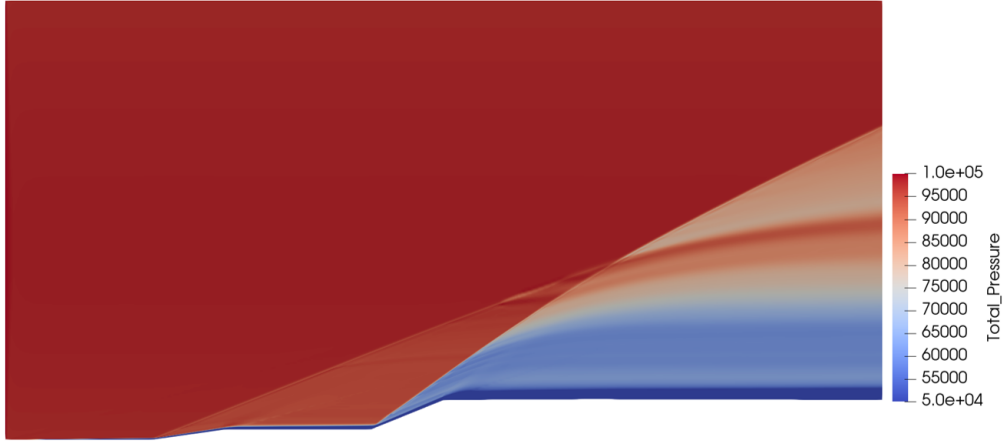


Figure 8: Contour plot of the total pressure (p_0) for the BSL EARSIM simulation

Since the shock angle of the first shock is smaller than that of the second, the two shocks interact with each other. However, this interaction is dominated by the second shock due to its greater strength. Downstream of the second shock, an expansion fan develops, which strongly bends the first shock and influences the position of the shock interaction. Finally, the shock system exits the domain through the zero-gradient outlet.

4.1 Inviscid Simulation

The results of the inviscid simulation reveal a double shock system with shock interaction and regions of high gradients. The Mach number after the first shock is measured as

$M_1 = 3.26$, and after the second shock as $M_2 = 2.31$ (see Table 4). In an inviscid simulation, no boundary layer develops; therefore, there is no influence from the boundary layer on the shock angles. This results in an underpredicted shock angle for β_1 and β_2 .

4.2 $k-\varepsilon$ Simulation

The $k-\varepsilon$ turbulence model was run on two different meshes to compare their accuracy and determine which one should be used for future investigations. The convergence behavior was clearly poorer when using the refined mesh (see Fig. 6). However, since the convergence criteria were still satisfied, the results could be analyzed.

The two simulations were compared based on the shock angles, total pressure values, and Mach number analysis. The results showed a similar flow behaviour through the first (weak) shock; however, a significant difference of 4.6% was observed in the total pressure after the second shock.

As previously discussed, a hexahedral structured mesh can reduce numerical diffusion if the cells are aligned with the flow direction. In the case of the universal grid, the cells are not aligned with the flow field, which causes the mesh to lose one of the main advantages of a structured grid and introduces numerical diffusion into the solution.

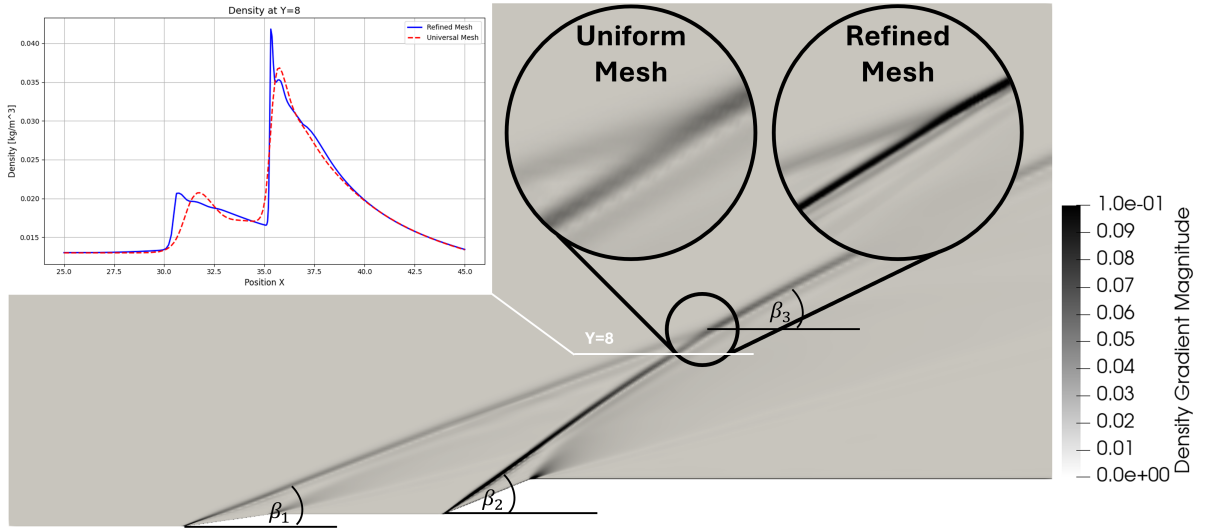


Figure 9: Density field compared between Universal- and Refined Mesh

The effect of numerical diffusion was investigated by visualizing the density field and analyzing the density distribution along a line between the points (25, 8) and (45, 8) (see fig. 9). The refined mesh shows well-defined peaks and strong gradients, which represent the expected physical behavior in the flow field. In contrast, the universal mesh exhibits a high dissipative error, resulting in the smoothing of peaks and damping of strong gradients.

From these observations, it can be concluded that the refined mesh provides more accurate results and minimizes numerical diffusion, making it the preferred choice for future investigations.

4.3 BSL EARSM Simulation

The BSL EARSM model predicts slightly smaller shock angles for the first shock ($\beta_1 = 21.54^\circ$) and a higher shock angle for the second shock ($\beta_1 = 35.05^\circ$). This can be explained by the dissipative nature of the $k-\varepsilon$ model. The shock angle after the shock interaction is also higher in the BSL EARSM case.

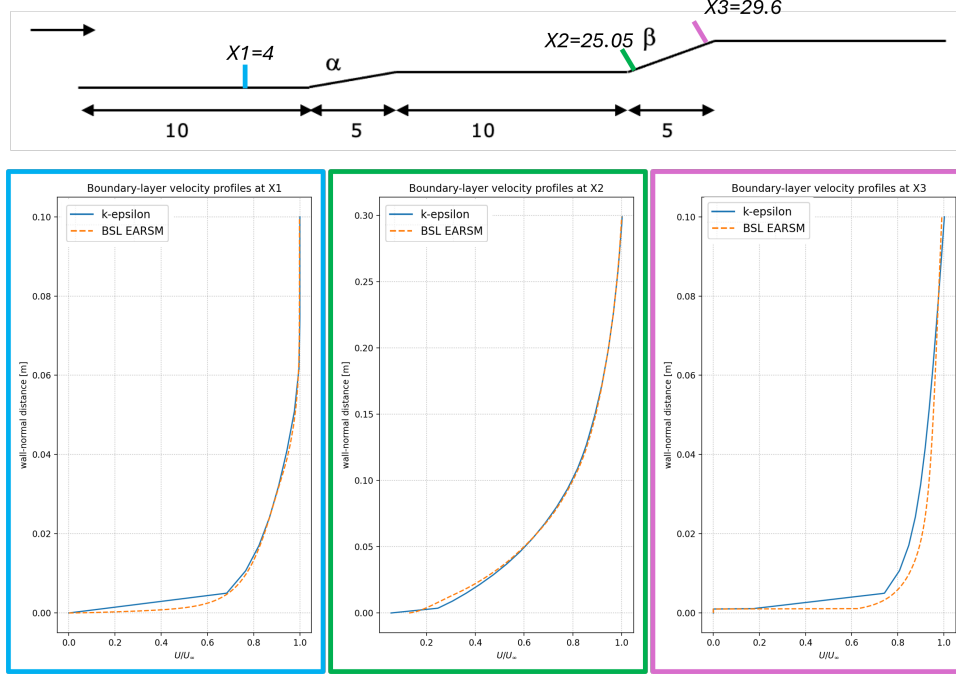


Figure 10: Boundary layer velocity profile plotted at three locations: close to the inlet (X1), at the beginning of the second ramp (X2), at the end of the second ramp (X3)

The influence of the boundary layer is also significant and can be investigated using the boundary layer velocity profiles. At position X1, the $k-\varepsilon$ model shows good agreement with the BSL EARSM velocity profile. The linear line in the $k-\varepsilon$ plot should not be misleading, as there is subgrid-scale modeling that matches the velocity profile of the wall model between the wall and the last resolved cell. A different behavior is observed when a pressure gradient is present. At position X2, close to the wall, the velocity is slightly overpredicted, while at position X3, the velocity is highly underpredicted in the $k-\varepsilon$ case compared to the BSL EARSM model. This demonstrates the limitations of wall modeling and highlights the major advantage of the BSL EARSM model in its ability to handle highly anisotropic flows.

Case	β_1	β_2	β_3	M_1	M_2	p_{01}	p_{02}
Inv.	20.41°	34.51°	26.47°	3.26	2.31	95 800 Pa	60 608 Pa
$k-\varepsilon$ (universal)	21.37°	35.41°	25.87°	3.26	2.36	96 300 Pa	63 960 Pa
$k-\varepsilon$ (refined)	21.64°	34.89°	26.48°	3.25	2.33	96 285 Pa	61 010 Pa
BSL EARSM	21.54°	35.05°	26.54°	3.20	2.14	93 883 Pa	60 488 Pa

Table 4: Comparison of key shock properties across simulations.



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1. File Report

Table 1. File Information for Inviscid_new_material_003

Case	Inviscid_new_material_003
File Path	C:/Users/balas/Documents/TUM/2025 SS/ACFD_Project_ramp/Final_sim/Final/Inviscid/Inviscid_new_material_003.res
File Date	04 September 2025
File Time	08:36:40 PM
File Type	CFX5
File Version	25.1

2. Mesh Report

Table 2. Mesh Information for Inviscid_new_material_003

Domain	Nodes	Elements
Default Domain	155648	77161

3. Physics Report

Table 3. Domain Physics for Inviscid_new_material_003

Domain - Default Domain	
Type	Fluid
Location	SOLID
<i>Materials</i>	
InviscidAir	
Fluid Definition	Material Library
Morphology	Continuous Fluid
<i>Settings</i>	
Buoyancy Model	Non Buoyant
Domain Motion	Stationary
Reference Pressure	0.0000e+00 [atm]
Heat Transfer Model	Total Energy
Include Viscous Work Term	True
Turbulence Model	Laminar

Table 4. Boundary Physics for Inviscid_new_material_003

Domain	Boundaries	
Default Domain	Boundary - Inlet	
	Type	INLET
	Location	INLET
	<i>Settings</i>	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Supersonic
	Heat Transfer	Total Temperature
	Total Temperature	9.0000e+02 [K]
	Mass And Momentum	Static and Total Pressure
	Relative Static Pressure	8.6600e+02 [Pa]
	Relative Total Pressure	1.0000e+00 [bar]
	Boundary - Outlet	
	Type	OUTLET
	Location	OUTLET
	<i>Settings</i>	
	Flow Regime	Supersonic
	Boundary - Side	
	Type	SYMMETRY
	Location	SIDES
	<i>Settings</i>	
	Boundary - Top	

	Type	SYMMETRY
	Location	TOP
	<i>Settings</i>	
	Boundary - Ramp	
	Type	WALL
	Location	WALL
	<i>Settings</i>	
	Heat Transfer	Adiabatic
	Mass And Momentum	Free Slip Wall

4. Solution Report

Table 5. Boundary Flows for Inviscid_new_material_003

Location	Type	Mass Flow	Momentum		
			X	Y	Z
Inlet	Boundary	4.5175e+02	5.4970e+05	-3.2142e-02	3.1987e-04
Outlet	Boundary	-4.5170e+02	-5.3899e+05	-2.6283e+04	7.7937e-11
Ramp	Boundary	0.0000e+00	-1.0672e+04	7.8251e+04	-7.1464e-03
Side	Boundary	0.0000e+00	2.3265e-13	2.3659e-12	6.8265e-03
Top	Boundary	0.0000e+00	-2.3731e-19	-5.1960e+04	8.5396e-10

Table 6. Forces and Torques for Inviscid_new_material_003

Location	Type	X	Y	Z
Ramp	Pressure Force	1.0672e+04	-7.8251e+04	7.1464e-03
	Viscous Force	0.0000e+00	0.0000e+00	0.0000e+00
	Total Force	1.0672e+04	-7.8251e+04	7.1464e-03
	Pressure Torque	3.9125e+04	5.3359e+03	-2.2775e+06
	Viscous Torque	0.0000e+00	0.0000e+00	0.0000e+00
	Total Torque	3.9125e+04	5.3359e+03	-2.2775e+06

5. User Data



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[5. User Data](#)

1. File Report

Table 1. File Information for k epsilon_uniform_mesh_probe_002

Case	k epsilon_uniform_mesh_probe_002
File Path	C:/Users/balas/Documents/TUM/2025 SS/ACFD_Project_ramp/Final_sim/Final/k-epsilon/k-epsilon_uniform_mesh_probe_002.res
File Date	05 September 2025
File Time	10:40:33 PM
File Type	CFX5
File Version	25.1

2. Mesh Report

Table 2. Mesh Information for k epsilon_uniform_mesh_probe_002

Domain	Nodes	Elements
Default Domain	155648	77161

3. Physics Report

Table 3. Domain Physics for k epsilon_uniform_mesh_probe_002

Domain - Default Domain	
Type	Fluid
Location	SOLID
<i>Materials</i>	
Air Ideal Gas	
Fluid Definition	Material Library
Morphology	Continuous Fluid
<i>Settings</i>	
Buoyancy Model	Non Buoyant
Domain Motion	Stationary
Reference Pressure	0.0000e+00 [atm]
Heat Transfer Model	Total Energy
Include Viscous Work Term	True
Turbulence Model	k epsilon
Turbulent Wall Functions	Scalable
High Speed Model	Off

Table 4. Boundary Physics for k epsilon_uniform_mesh_probe_002

Domain	Boundaries	
Default Domain	Boundary - Inlet	
	Type	INLET
	Location	INLET
	<i>Settings</i>	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Supersonic
	Heat Transfer	Total Temperature
	Total Temperature	9.0000e+02 [K]
	Mass And Momentum	Static and Total Pressure
	Relative Static Pressure	8.6600e+02 [Pa]
	Relative Total Pressure	1.0000e+00 [bar]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio
	Boundary - Outlet	
	Type	OUTLET
	Location	OUTLET
	<i>Settings</i>	
	Flow Regime	Supersonic
	Boundary - Side	
	Type	SYMMETRY

	Location	SIDES
	<i>Settings</i>	
	Boundary - Top	
	Type	SYMMETRY
	Location	TOP
	<i>Settings</i>	
	Boundary - Ramp	
	Type	WALL
	Location	WALL
	<i>Settings</i>	
	Heat Transfer	Adiabatic
	Mass And Momentum	No Slip Wall
	Wall Roughness	Smooth Wall

4. Solution Report

Table 5. Boundary Flows for k epsilon_uniform_mesh_probe_002

Location	Type	Mass Flow	Momentum		
			X	Y	Z
Inlet	Boundary	4.5280e+02	5.5066e+05	-3.7415e-02	3.2043e-04
Outlet	Boundary	-4.5282e+02	-5.3956e+05	-2.7267e+04	-5.1165e-08
Ramp	Boundary	0.0000e+00	-1.1120e+04	7.9219e+04	-7.3668e-03
Side	Boundary	0.0000e+00	2.3306e-13	2.3785e-12	7.0464e-03
Top	Boundary	0.0000e+00	-1.6160e-22	-5.1960e+04	8.5396e-10

Table 6. Forces and Torques for k epsilon_uniform_mesh_probe_002

Location	Type	X	Y	Z
Ramp	Pressure Force	1.0469e+04	-7.9278e+04	7.3667e-03
	Viscous Force	6.5119e+02	5.9510e+01	3.4380e-08
	Total Force	1.1120e+04	-7.9219e+04	7.3668e-03
	Pressure Torque	3.9639e+04	5.2343e+03	-2.3072e+06
	Viscous Torque	-2.9755e+01	3.2560e+02	5.1291e+02
	Total Torque	3.9609e+04	5.5599e+03	-2.3067e+06

5. User Data



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1. File Report

Table 1. File Information for k_epsilon_refined_1_001

Case	k_epsilon_refined_1_001
File Path	C:/Users/balas/Documents/TUM/2025 SS/ACFD_Project_ramp/Final_sim/Final/k-epsilon/post/k-epsilon_refined_1_001.res
File Date	04 September 2025
File Time	08:03:14 PM
File Type	CFX5
File Version	25.1

2. Mesh Report

Table 2. Mesh Information for k epsilon_refined_1_001

Domain	Nodes	Elements
Default Domain	121570	60289

3. Physics Report

Table 3. Domain Physics for k epsilon_refined_1_001

Domain - Default Domain	
Type	Fluid
Location	SOLID
<i>Materials</i>	
Air Ideal Gas	
Fluid Definition	Material Library
Morphology	Continuous Fluid
<i>Settings</i>	
Buoyancy Model	Non Buoyant
Domain Motion	Stationary
Reference Pressure	0.0000e+00 [atm]
Heat Transfer Model	Total Energy
Include Viscous Work Term	True
Turbulence Model	k epsilon
Turbulent Wall Functions	Scalable
High Speed Model	Off

Table 4. Boundary Physics for k epsilon_refined_1_001

Domain	Boundaries	
Default Domain	Boundary - Inlet	
	Type	INLET
	Location	INLET
	<i>Settings</i>	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Supersonic
	Heat Transfer	Total Temperature
	Total Temperature	9.0000e+02 [K]
	Mass And Momentum	Static and Total Pressure
	Relative Static Pressure	8.6600e+02 [Pa]
	Relative Total Pressure	1.0000e+00 [bar]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio
	Boundary - Outlet	
	Type	OUTLET
	Location	OUTLET
	<i>Settings</i>	
	Flow Regime	Supersonic
	Boundary - Side	
	Type	SYMMETRY

	Location	SIDES
	<i>Settings</i>	
	Boundary - Top	
	Type	SYMMETRY
	Location	TOP
	<i>Settings</i>	
	Boundary - Ramp	
	Type	WALL
	Location	WALL
	<i>Settings</i>	
	Heat Transfer	Adiabatic
	Mass And Momentum	No Slip Wall
	Wall Roughness	Smooth Wall

4. Solution Report

Table 5. Boundary Flows for k epsilon_refined_1_001

Location	Type	Mass Flow	Momentum		
			X	Y	Z
Inlet	Boundary	4.5280e+02	5.5066e+05	-2.7071e-02	1.0124e-01
Outlet	Boundary	-4.5280e+02	-5.3947e+05	-2.7397e+04	-1.3193e-02
Ramp	Boundary	0.0000e+00	-1.1191e+04	7.9427e+04	1.3492e-03
Side	Boundary	0.0000e+00	4.9882e-13	3.4127e-13	-8.1124e-02
Top	Boundary	0.0000e+00	8.2669e-04	-5.2029e+04	-8.2705e-03

Table 6. Forces and Torques for k epsilon_refined_1_001

Location	Type	X	Y	Z
Ramp	Pressure Force	1.0545e+04	-7.9487e+04	-1.3492e-03
	Viscous Force	6.4575e+02	6.0218e+01	-4.3131e-08
	Total Force	1.1191e+04	-7.9427e+04	-1.3492e-03
	Pressure Torque	3.9744e+04	5.2726e+03	-2.3034e+06
	Viscous Torque	-3.0108e+01	3.2288e+02	5.3434e+02
	Total Torque	3.9714e+04	5.5954e+03	-2.3029e+06

5. User Data



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[5. User Data](#)

1. File Report

Table 1. File Information for BSL_EARSM_refined_prism_layer_002

Case	BSL_EARSM_refined_prism_layer_002
File Path	C:/Users/balas/Documents/TUM/2025 SS/ACFD_Project_ramp/Final_sim/Final/BSL_EARSM/Post/BSL_EARSM_refined_prism_layer_002.res
File Date	04 September 2025
File Time	07:44:42 PM
File Type	CFX5
File Version	25.1

2. Mesh Report

Table 2. Mesh Information for BSL_EARSM_refined_prism_layer_002

Domain	Nodes	Elements
Default Domain	141148	70039

3. Physics Report

Table 3. Domain Physics for BSL_EARSM_refined_prism_layer_002

Domain - Default Domain	
Type	Fluid
Location	SOLID
<i>Materials</i>	
Air Ideal Gas	
Fluid Definition	Material Library
Morphology	Continuous Fluid
<i>Settings</i>	
Buoyancy Model	Non Buoyant
Domain Motion	Stationary
Reference Pressure	0.0000e+00 [atm]
Heat Transfer Model	Total Energy
Include Viscous Work Term	True
Turbulence Model	BSL EARSM
Turbulent Wall Functions	Automatic
High Speed Model	Off

Table 4. Boundary Physics for BSL_EARSM_refined_prism_layer_002

Domain	Boundaries	
Default Domain	Boundary - Inlet	
	Type	INLET
	Location	INLET
	<i>Settings</i>	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Supersonic
	Heat Transfer	Total Temperature
	Total Temperature	9.0000e+02 [K]
	Mass And Momentum	Static and Total Pressure
	Relative Static Pressure	8.6600e+02 [Pa]
	Relative Total Pressure	1.0000e+00 [bar]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio
	Boundary - Outlet	
	Type	OUTLET
	Location	OUTLET
	<i>Settings</i>	
	Flow Regime	Supersonic
	Boundary - Side	
	Type	SYMMETRY
	Location	SIDES
	<i>Settings</i>	
	Boundary - Top	
	Type	SYMMETRY
	Location	TOP
	<i>Settings</i>	
	Boundary - Ramp	

	Type	WALL
	Location	WALL
	<i>Settings</i>	
	Heat Transfer	Adiabatic
	Mass And Momentum	No Slip Wall
	Wall Roughness	Smooth Wall

4. Solution Report

Table 5. Boundary Flows for BSL_EARSM_refined_prism_layer_002

Location	Type	Mass Flow	Momentum		
			X	Y	Z
Inlet	Boundary	4.5280e+02	5.5020e+05	-4.1419e-02	1.0115e-01
Outlet	Boundary	-4.5280e+02	-5.3898e+05	-2.7383e+04	-1.3299e-02
Ramp	Boundary	0.0000e+00	-1.1220e+04	7.9884e+04	1.3623e-03
Side	Boundary	0.0000e+00	4.8847e-13	3.6712e-13	-8.0842e-02
Top	Boundary	0.0000e+00	8.3047e-04	-5.2502e+04	-8.3668e-03

Table 6. Forces and Torques for BSL_EARSM_refined_prism_layer_002

Location	Type	X	Y	Z
Ramp	Pressure Force	1.0567e+04	-7.9944e+04	-1.3624e-03
	Viscous Force	6.5314e+02	5.9571e+01	3.0051e-08
	Total Force	1.1220e+04	-7.9884e+04	-1.3623e-03
	Pressure Torque	3.9972e+04	5.2835e+03	-2.3163e+06
	Viscous Torque	-2.9785e+01	3.2656e+02	4.9381e+02
	Total Torque	3.9942e+04	5.6101e+03	-2.3159e+06

5. User Data